

Advanced Audio Coding

WMA and RealAudio 8 may have the advantage over MP3: Their files are half the size at CD-quality rates. However, a newer standard called Advanced Audio Coding (AAC) may give them a run for their money. AAC was co-developed by the Fraunhofer IIS-A, Dolby, and AT&T, and it is part of the MPEG-2 specification. AAC's algorithm is different from MP3's, and AAC incorporates additional features that boost the efficiency of encoding. Basically, AAC delivers better-quality audio with 30 per cent less data than MP3. AAC is widely known to be the best when it comes to audio compression, and a number of desktop players, like the new MusicMatch Jukebox 6.0, are beginning to incorporate it into the suite of codecs they support. As of today, there are no portable players that support AAC, but with its upgradable firmware, it may just be a matter of time before AAC becomes as common as MP3.

kbps to a lower value whenever possible (if there is not much audio going on) and can accelerate the bit rate by 32 kbps as and when required. Essentially, VBR is used to provide for an efficient compromise between file size and audio quality. Whenever a silent period is encountered during a track, the bit rate can be decreased, thus conserving on the size of that particular segment. Whenever higher quality is in need, the bit rate can increase giving a fuller, more accurate reproduction of the original track, but at the sacrifice of space.

The problem encountered with VBR is that as the bit rate is constantly changing from one segment to another, several players have a problem in determining correct track length. Though a standard exists where track length is recorded into the first segment of an MP3 file, not all encoders can do this.

A negative point from the perspective of the recording industry with the MP3 format is that there are no security features that have been included in the format—it is free for everyone to pirate and share. As for the licensing part of MP3, a company needs to pay \$3.75 for each device capable of playing back the MP3 format.

Emerging standards

Though the MP3 format does a satisfactory job in compressing song data and maintaining a high standard of audio quality, recent adven- tures in technology permit even further compression of song data. Thompson Technologies, the co-creators of the MP3 format along with the Fraunhofer Institute, have teamed up with Coding Technologies to produce the next-generation MP3 format called MP3Pro. MP3Pro operates in a similar fashion to the MP3 Codec with a few enhancements. A codec is analogous to a device driver in that it allows playback of a particular audio format using specified parameters. MP3Pro is completely backward compatible with the original MP3 standard but you will not reap the full benefits of this codec unless the suitable codec or plugin is installed on the computer.

In MP3Pro, the encoder splits up the encoding into two main parts: the low frequency range

and the high frequency range. The low frequency range consists of signals below 10 KHz while the high frequency range lies above 10 KHz. This technique proves to be a more efficient way of encoding sound into MP3Pro format and also contributes to increasing overall sound quality.

One problem encountered with the MP3 format is that at low bit rates, MP3 files run out of bits to compress the music in full audio bandwidth. That is why at bit rates of around 48 kbps to 64 kbps, MP3 music begins to sound dull as it lacks a good deal of high frequency signals. At 64 kbps, MP3 can either produce an audio bandwidth of approximately 10 KHz or introduce coding artifacts. Coding artifacts are used to compensate for the shortage of bits while sampling at low bit rates in order to attempt to represent the complete audio bandwidth.

MP3Pro is a result of combining two components: MP3 and Spectral Band Replication, or SBR. SBR is a new coding technology which offers FM-quality audio (15 KHz) by increasing audio bandwidth of a standard MP3 sample.

SBR is a new method used in conjunction with the MP3 format to reproduce high frequencies in low bit rate encoded audio files. When used with SBR, the underlying MP3 encoder is responsible for transmitting frequencies below 10 KHz while SBR reconstructs all frequencies above the 10 KHz range. This is a process which occurs during playback of the MP3Pro file. First, the MP3 decoder processes the file (and frequencies below 10 KHz), then it transmits the decoded portion to SBR for analysis and high frequency enhancement of the decoded portion. In effect, the SBR accurately represents the original music before the encoding process began. This reconstruction is not only accurate in the frequency domain, but in the time domain as well. The end result is full bandwidth audio decoding at low data rates and increased compression efficiency.

The free version of MP3Pro supports a 64 kbps bit rate and a maximum sampling rate of 44.1 KHz. The official full version of the codec will allow a 96 kbps bit rate, which easily rivals standard MP3 files encoded at 128 kbps in terms of audio reproduction. Like MP3, MP3Pro does

